# THE RIGHT-LOCALIZED PULLBACK OF A CATEGORY FIBRED IN GROUPOIDS IS FIBRED IN GROUPOIDS

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*Editorial note.* No omission markers in-span; proof complete as written.

*Allowed standard prerequisites (editorial).*

- Pullback of a fibred category. lemma-fibred-category-pullback (same chapter, named). Provides that u_pS is a fibred category and the fraction description of its morphisms.
- Fibred in groupoids. categories-lemma-fibred-groupoids (cross-chapter, named). Reduces the groupoid condition to invertibility of morphisms over identity arrows, and gives that all morphisms of S are strongly cartesian.

*Source reference.* Original reference 003V: focused-seeds/geometry/S9-stk-fibred-groupoids-category-pullback/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 04WG: sources/stacks-stacks/pinned-original-source.tex, source lines 3188–3188. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *With notation and assumptions as in Lemma 04WG. If $\mathcal{S}$ is fibred in groupoids, then $u_p\mathcal{S}$ is fibred in groupoids.*

*Proof.* By Lemma 04WG we know that $u_p\mathcal{S}$ is a fibred category. Let $f : X \to Y$ be a morphism of $u_p\mathcal{S}$ with $p_p(f) = \mathrm{id}_V$. We are done if we can show that $f$ is invertible, see Categories, Lemma 003V. Write $f$ as the equivalence class of a pair $((a, b, \alpha), r)$ with $r \in R$. Then $p_p(r) = \mathrm{id}_V$, hence $p_{pp}((a, b, \alpha)) = \mathrm{id}_V$. Hence $b = \mathrm{id}_V$. But any morphism of $\mathcal{S}$ is strongly cartesian, see Categories, Lemma 003V hence we see that $(a, b, \alpha) \in R$ is invertible in $u_p\mathcal{S}$ as desired. □

*Date*: stacks-project:stacks.tex:local-snapshot-20261012; GNU FDL 1.2 or later.